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Development of a MALTA2-Based Fast MAPS Tracker: Detector Design, Readout Firmware, and Performance Validation

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Abstract To advance the forward tracking capability at the future Electron-Ion Collider (EIC), a MALTA2-based Fast Monolithic Active Pixel Sensor (MAPS) Tracker is proposed to deliver better than $7\mu m$ spatial resolution, better than 2ns readout window and high radiation tolerance.

1 Introduction - (lead by Xuan, Dien)

The Electron-Ion Collider (EIC) will provide high-luminosity, high-energy electron+proton ($e+p$) and electron+nucleus (e+A) collisions, enabling three-dimensional imaging of nucleons and nuclei, advancing our understanding of the proton spin puzzle, and investigating the origin of visible matter as well as the machanisms of hadron formation [1]. The EIC is designed to accommodate up to two Interaction Points (IPs): one for the project detector being developed by the ePIC collaboration and another reserved for a future second EIC detector. The collider will operate with a bunch crossing interval of 10.2 ns, while the Deep Inelastic Scattering (DIS) physics event rate is expected to be approximately 500 kHz rate. Consequently, the detector systems will be exposed to background particle rates that are substantially higher than those associated with DIS physics events, presenting significant challenges for detector design, readout, and event reconstruction. Recent experimental and theoretical studies for the EIC [2–7] have demonstrated that precise reconstruction of heavy-flavor hadrons and jets in the forward pseudorapidity region ($\eta > 2$) is essential for constraining nucleon and nuclear parton distribution functions (PDFs) at high Bjorken-x (x_{BJ}), as well as for elucidating heavy-quark hadronization mechanisms. These measurements are directly aligned with the primary scientific objectives of the EIC [1].

To support either a future upgrade of the ePIC detector or the second detector at the EIC, we are developing a forward silicon tracker based on the Depleted Monolithic Active Pixel Sensor (DMAPS) technology, specifically the MALTA2 sensor [8]: Fast MAPS Tracker (FMT). A comprehensive detector R&D program, together with detailed simulation studies, has been carried out to evaluate the detector performance and optimize its design. The primary objectives of this detector are: 1) to achieve fine spatial resolution and fast timing capability while maintaining reliable performance under the radiation doses from multiple years of EIC operation; 2) to improve charged-particle tracking performance in the forward pseudorapidity region; and 3) to provide sufficient time resolution to distinguish particles originating from consecutive bunch crossings separated by 10.2 ns, thereby mitigating beam-induced background contamination in physics events.

In this paper, we present the design and development of the fundamental building unit of the FMT, the quad-sensor module, together with updates to the readout firmware relative to the previous sensor readout system. We then present detailed simulation studies evaluating the detector's tracking performance, timing capability, and impact from EIC backgrounds in the forward region. These results provide valuable guidance for the design optimization of a future forward tracking detector at the EIC.

2 MALTA2 Technology - (Lead by Carlos)

The Tower 180 nm CMOS Imaging Sensor (CIS) technology, implemented in the MALTA family of Depleted Monolithic Active Pixel Sensors (DMAPS) [8, 9], offers several

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advantages that make it an attractive candidate for tracking detectors at the EIC. First, its monolithic architecture integrates the sensing element and front-end readout electronics on the same silicon substrate, eliminating the need for costly and material-intensive bump-bonding. This integration reduces both the detector cost and material budget. Second, this technology empolys small collection electrodes with an input capacitance below 5 fF, resulting in low noise (< 20 ENC), low power consumption (< 20 mW/cm^2), and fast signal formation with a response time below the EIC bunch-crossing interval of 10.2 ns.

The MALTA2 sensor used in this work is the second-generation device in the MALTA sensor family, with significant improvements in in-pixel charge collection efficiency and radiation tolerance. Each MALTA2 sensor comprises a matrix of 224×512 pixels, corresponding to an active area of 18.3 mm^2, and with a pixel pitch of 36.4 μm. The sensor integrates an analog front-end within each pixel that functions as a charge amplifier and discriminator prior to digitization. A hierarchical asynchronous data-merging architecture enables simultaneous transmission of hit information from across the entire matrix, providing fast readout and enhanced timing resolution.

The MALTA series sensors have demonstrated excellent performance in beam test [10], including fine spatial granularity, a time resolution better than 2 ns, a signal-to-noise ratio of approximately 20 for minimum-ionizing particle (MIP) signals, resulting in a low fake-hit rate, and high charge-collection efficiency. These characteristics are critical for precise vertex and tracking reconstruction in the high-rate EIC environment. Furthermore, the high-resistivity, fully depleted sensing volume enables efficient charge collection and good radiation tolerance, while the thin sensor design reduces multiple scattering, thereby improving both momentum and impact-parameter resolutions. Collectively, these features make the Tower 180 nm CMOS technology a compelling choice for constructing large-area, high-performance, and cost-effective silicon tracking systems for future EIC experiments.

3 Detector and Module Design - (lead by Lindsey, Kerem)

3.1 FMT geometry

To improve tracking performance in the forward pseudo-rapidity region ($\eta > 2.5$), the current Fast MAPS Tracker (FMT) design consists of two silicon tracking disks surrounding the beam pipe and positioned downstream of the central tracking system. The detector geometry has been optimized by balancing forward tracking acceptance with detector integration constraints. The longitudinal position of the first disk was optimized within the range of 140-150 cm from the interaction point, while the second disk was placed between 160 and 165 cm. The ePIC interaction region employs a 25 mrad beam-crossing angle, whereas the interaction region for the future second EIC detector has not yet been finalized. Consequently, the inner radius of the FMT is constrained by the current ePIC beam-pipe design. In the present design, both tracking disks have an inner radius of 7.012 cm and the outer radius of 23.012 cm. Fig 1 presents the geometry of the FMT implemented in the standalone simulation. The first and second detector disks are positioned at z = 145 cm and z = 165cm, respectively.

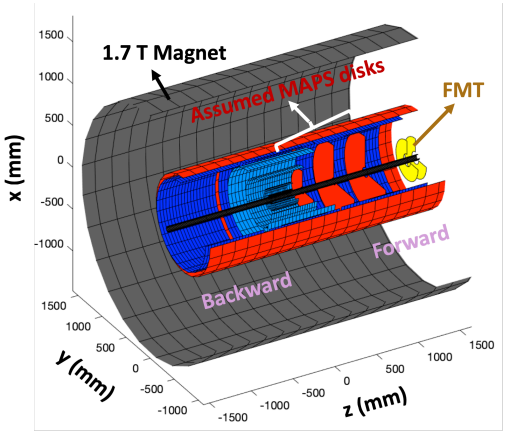


Fig. 1 Standalone simulation geometry showing the Fast MAPS Tracker (FMT, yellow) positioned downstream of the three MAPS tracking disks (red) within the ePIC solenoidal magnet (grey).

3.2 MALTA2 quad-sensor module prototype design

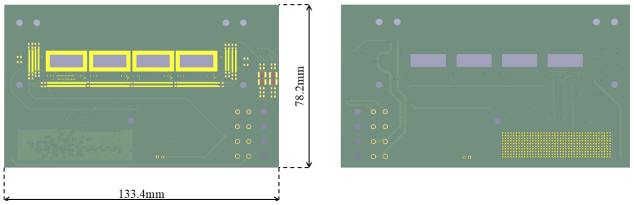


Fig. 2 3D rendering of FMT MATLA2 Quad Module FPC prototype. The front view is shown in the left and the back view is shown in the right.

The MALTA2 quad-sensor module is designed as the fundamental building block of the FMT. As shown in Fig 2, four MALTA2 sensors are mounted side by side on the prototype Flexible Printed Circuit (FPC). The readout and power pads are located along both sides of the MALTA2, enabling the interconnection of multiple sensors to construct a larger detector matrix while providing flexibility in the placement of the master chip. In the slave-master readout architecture,

all detector data are transmitted to and read out through the master chip. The master chip collects data from the upstream slave chips and outputs the aggregated data stream through the Low-Voltage Differential Signaling (LVDS) pads located on its bottom side, as seen in Fig 3. This modular design can minimize inter-chip communication distances while requiring only a single high-speed output link for the readout of the entire detector matrix. In the prototype carrier board used in this study, three slave chips are arranged in a row adjacent to the master chip, which is positioned on the left side of the module.

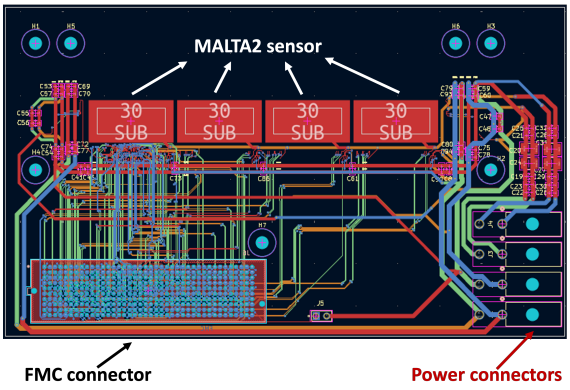


Fig. 3 Technical layout of the MALTA2 quad-sensor module prototype. The diagram illustrates the geometric arrangement of the four MALTA2 sensors, wire-bonding pads, and the routing paths for power distribution and high-speed differential signals.

The MALTA2 sensors are integrated onto a custom-designed FPC that provides the mechanical support and electrical interfaces required for tracker operation. Owing to the small bond-pad dimensions (88 $\mu m \times 88$ μm) and fine pad pitch on both the sensors and the FPC, reliable chip-to-chip and chip-to-board interconnections are critical for the assembly of the module. The FPC layout was optimized to ensure precise alignment of the electrical and mechanical components. The MALTA2 sensors were attached to the FPC using aluminum wedge wire bonding performed at the CERN Bonding Laboratory. Figure 4 illustrates the wire-bonding scheme required for sensor operation. In this prototype, the leftmost sensor serves as the master chip, through which all detector readout and slow-control communication are routed, as reflected by the high density of wire bonds along its bottom edge.

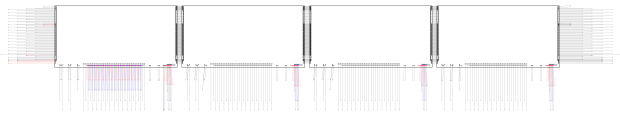


Fig. 4 Schematic of the wire-bonding layout for the MALTA2 quad-sensor module.

3.3 Initial Quad Module Testing

4 MALTA2 quad-sensor module readout architecture and firmware development - (lead by Joellen, Kerem)

4.1 Readout Architecture

The MALTA2 quad-sensor module utilizes 38 Low-Voltage Differential Signaling (LVDS) pairs originating from the bottom side of the master chip. Each MALTA2 sensor comprises of 256 double column of pixels, which form the basis of the hierarchical data-merging architecture within the pixel matrix. When a hit is registered, the corresponding data propagates toward the end of the column as a 22-bit word, where the data-merging process begins. A hierarchy of eight merging stages consolidates the hit information from all 256 double columns into a 37-bit output word for transmission through the LVDS links, including a stage that combines the positive- and negative-parity groups within each double column. CMOS transistors located along the left and right edges of each chip enable data to be forwarded from the three secondary chips to the master chip, where the data streams are combined for final readout. The propagation delays associated with transferring data from the secondary sensors to the master chip are accounted for through the appropriate timing calibration/synchronization procedure. After the successive inter-chip merging stages are completed, the resulting data are transmitted through the LVDS links to the FPGA for further processing.

4.2 Firmware Architecture

The readout of the MALTA2 quad-module is being developed using the Alinx AXKU040 (Kintex UltraScale) platform. The custom FPGA design is based on a previous version that used the KC705 (Kintex 7) platform for readout of a single MALTA2 sensor. Firmware updates were introduced to support quad-module operation by extending the data-input and slow-control interfaces while retaining the existing asynchronous readout architecture. This upgrade also preserved the established DAQ-side behavior, including the sampling rate, timing conventions, register-level operation, and output data format, to continue utilizing existing DAQ-side software.

For the quad-module implementation, the FPGA receives 37 differential signal pairs from the module. Because of the

bus sharing for chip readout, additional information is needed to distinguish which of the four chips the data originates from. A two-bit chip identifier is associated with the data stream, with the values 00, 01, 10, and 11 identifying the four individual sensors. This allows data from all four sensors to be transmitted through the common module readout path while preserving the identity of the sensor from which each hit originated.

The readout of the chips is fully asynchronous with no clock distribution needed between the chips and FPGA. Each half of the differential signal pairs is asynchronously oversampled using separate Input Delay Elements (IDELAYE) and Input Serializer/Deserializers (ISERDESE). Using 640 MHz DDR sampling together with the independently delayed P and N signal paths, the implemented oversampling scheme achieves an effective sampling rate of 2.56 Gb/s.

Migration from the Kintex 7 platform to the Kintex UltraScale platform required updating the asynchronous sampling module to use IDELAYE3/ISERDESE3 elements instead of IDELAYE2/ISERDESE2 elements. While the IDELAYE3 primitives provide more tap delays than the IDELAYE2 primitives, the finer delay steps are not exposed to software to maintain compatibility with the existing DAQ software interface. Instead, each software delay step maps to 16 steps in the FPGA. The internal timing paths were also revised to account for changes in the UltraScale delay, clocking, and deserialization resources while still retaining the same software interface for calibration and outputs.

Each MALTA2 sensor is configured through its dedicated slow-control interface, in which the configuration data is serially shifted into the on-chip configuration registers. For quad-module operation, separate slow-control data paths are provided for the four sensors, allowing the configuration of each chip to be programmed independently. This enables independent adjustment of the chip-level configuration parameters, including the programmable DAC settings, for each MALTA2 sensor.

5 Performance Validation in Simulation - (Lead by Abhishek, Amelia, Lucian, Vlad)

As discussed in section (ref section 3.1) and shown in Fig.3 (ref fig2) a quad module made of four MALTA2 sensors in series is simulated in GEANT4. Each sensor is identical and placed horizontally adjacent to each other, as shown in Fig. 6. Each MALTA2 sensor is implemented in a data-driven fast simulation method. A detailed characterization and implementation of this fast simulation approach is described in Ref. [11]. This study primararily focuses on the abilty of reconstruction of signals (pions) by the MALTA2 sensors in the presence of background particles, mostly electrons and positrons, that are expected to be present near the beam-pipe of EIC. A detailed study of the background rate and its impact on the reconstruction efficiency of the MALTA2 sensors is presented in the following subsections.

5.1 Telescope geometry and setup in simulation

Each quad module consists of four MALTA2 sensors placed side by side in the *xy* plane as shown in Fig. 6 with negligible gap of $\approx 1 \mu m$. The sensor positions are listed in Table 1. A 29 μm thick silicon layer is used to construct sensor and it is defined as senstive detector volume in GEANT4. Both Elecromagnetism and Hadronic physics list are being used here in this simulation. A particle gun is used to generate signal particles in a longitudinal direction and it is centered at (18.64, 0.0, -1000.0) mm.

$$D^0 \to K^- \pi^+, \qquad \bar{D}^0 \to K^+ \pi^-. \tag{1}$$

Positive charged Pions are used as signal particles since they are decay-products of D^0 as shown in Eq. 1 and that is one of benchmark for track reconstruction at EIC. (we need more rigrous justification) Pions (signal) are generated from a three-dimensional Gaussian distribution, with standard deviations of 0.5 mm in *x* and *y* and 100 mm in *z*. The generated vertex distribution is shown in Fig. 7. These particles deposits energy in the sensor. The two-dimensional distribution of raw hits is shown in Fig. 8 and deposited energy will modeled as charge collection and that is simulated using the data-driven fast simulation method described in Ref. [11]. Along with charge collection, a paramteric model for time-walk is used here to simulate realistic MALTA2 sensor timing and it is explained in Ref. [11].

MALTA Readout: An asynchronous readout architecture is used for storing hit information. Sensor is divided into 512 x 512 pixels and those pixels are further grouped into groups and rows.

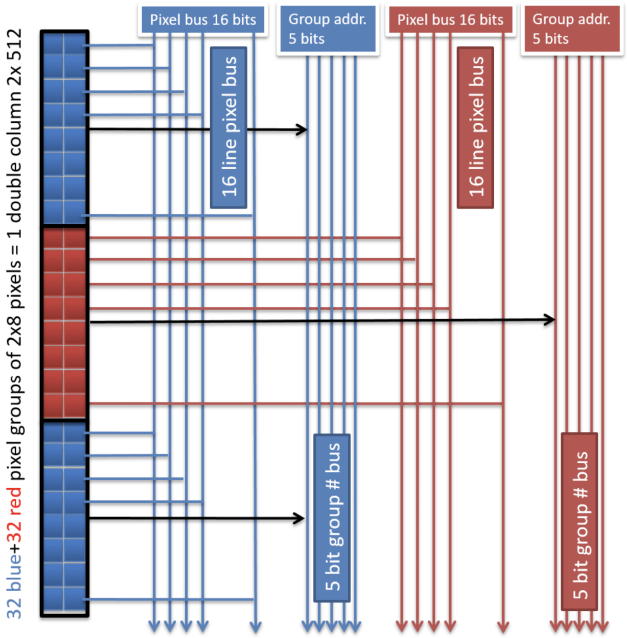


Fig. 5 MALTA sensor: the 512×512 pixels are grouped into columns and rows. The asynchronous readout architecture is described in https://cds.cern.ch/record/2691881/files/10.1088_1748-0221_14_06_C06019.pdf.

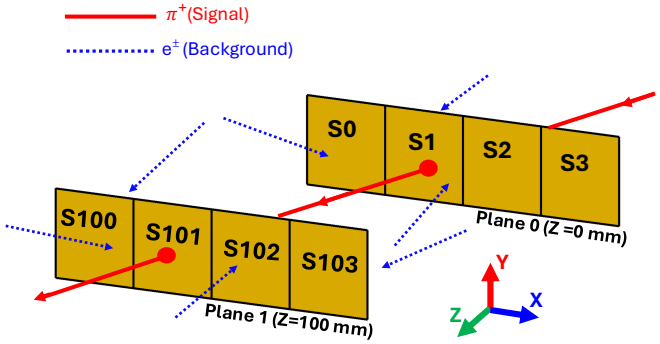


Fig. 6 MALTA QUAD Module geometry: eight 18.6×18.6 mm^2 silicon sensors in two planes separated by 100 mm. Red: 1 GeV π^+ signal beam generated at $z = -1000$ mm and focused on the second sensor (S1 & S101) of each plane. Blue dashed: randomly generated background $e^{\pm}$.

Table 1 Positions of the eight MALTA2 sensors in the quad-module geometry. All sensors lie in the $y = 0$ plane and are placed side by side along x.

Plane	z [mm]	Sensor x-centre [mm]			
		S0	S1	S2	S3
0	0	0.00	18.64	37.27	55.91
1	100	0.00	18.64	37.27	55.91

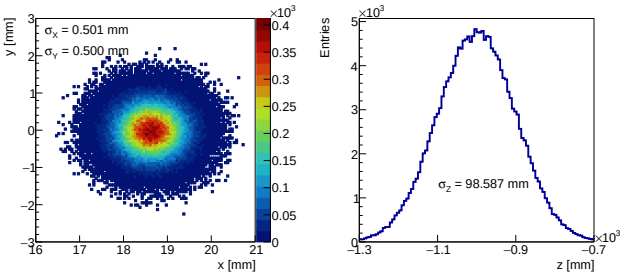


Fig. 7 Generated signal-beam vertex positions: (left) transverse (x, y) distribution and (right) longitudinal z distribution.

The collected charge is digitized by enconding hits into 32-Bit words and merging withing 1.6 ns window and later decoded to get digtized infor. The raw hit and digitized information is stored in different ROOT files for further analysis. Signal beam has a veto of 2 µs (cite Xuan arxiv paper) this is taken from the fact that EIC has bunch-crossing of 10.2 ns but signal only produces every 2 µs. The energy of signal These signal particles are reconstructed by the sensors in both planes and those hits are matched with generated truth tracks both spatially and temporarily window. If a Hit (X,Y) on sensor is with in radius of 100 μm and within 500 *ns* time window with refernce to generated truth. It will be matched with generated truth and defined as Hit matching. We define a hit efficiency as the ratio of the number of signal particles that are reconstructed by the sensor to the total number of signal particles generated in the simulation. Without any bcakground and signal hit efficiency for 1 GeV Pion is $\approx$ 96.8%. The efficiency here is lost in two component firsr major compnent is pion's short life-time. A 1 GeV Pion has probability of decay $\approx$ 1.6%. Second compoenent is charge sharing among 4 pixels and a signal below threshold of 150 e- can be lost. If Pion life-time set infinite in simulation ist Hit efficiency is $\approx$ 99% for the plane 0.

5.2 Time Resolution (no background)

Plot: Time resolution (compare: without background, with 1,2,3 particles per event) Validated with data.

5.3 Background rate

Explain the expected background at the location of the FMT (from Amelia's plots?):

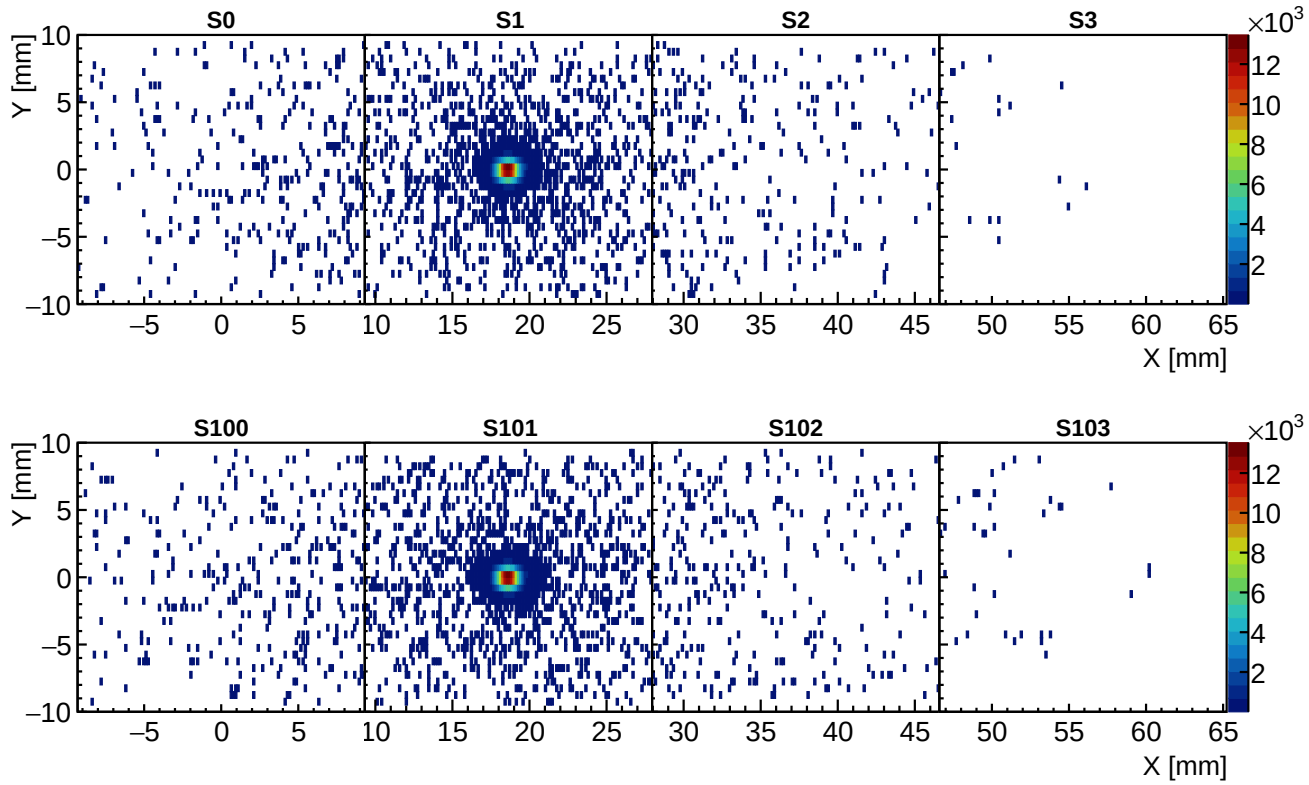


Fig. 8 Two-dimensional distribution of raw hits in the eight MALTA2 sensors. The top row shows sensors S0–S3 of the first plane (0) and the bottom row sensors S100–S103 of the second plane (1).

- Select region of $1 \times 2\,\mathrm{cm}^2$ with highest hit rate.
- Show distribution of hits per 2 µs event
- Show PDG code of background particles. Is it primarily photons and electrons/ positrons?

Explain the simplifications/ assumptions that go into the MALTA2 simulation.

- Identify maximum expected background hit rate per sampled event.
- Spatial distribution is approximately flat within individual MALTA2 sensor.

5.4 Sampling of EIC background

The momentum magnitude distributions of the background electrons and positrons are each modelled by a Crystal Ball function. The fitted parameters are used to sample the background momentum in every 2 µs window via inverse-transform sampling. The corresponding fit for electrons is shown in Fig. 9. Background vertices are generated uniformly within a three-dimensional box spanning the simulation world volume. The direction of each background particle is chosen such that it always crosses one of the two detector planes, as shown in Fig. 6. After fixing momentum and random direction rate of background particle is sampled using poisson dstrubtuion. EIC background rate is $\approx$ 1.5 particles per 2 μs.

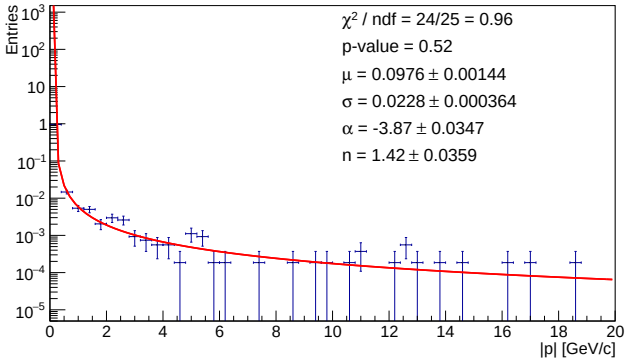


Fig. 9 Crystal Ball fit to the momentum magnitude distribution of background electrons.

5.5 Hit efficiency with background

The background rate is varied from zero to 15.84 particles per 2 µs event, and the hit efficiency is measured at each rate, as shown in Fig. 10. The per-plane hit efficiency is defined as the fraction of signal pions that passed the Hit-matching critires defined in section 5.1 in a given plane.

Plane 0 and plane 1 follow the same slow decrease with the background rate, with plane 1 systematically lower by about 1% because a fraction of the pions scattered between the two planes. The coincidence efficiency requires a reconstructed hit in both planes within the coincidence window, and is therefore lower than either per-plane efficiency. The coincidence efficiency decreases faster than the per-plane efficiencies because the signal pion is scattered in the first sensor layer, which displaces its hit position in the second plane and causes a small fraction of events to fail the coincidence requirement as the background occupancy increases. In principle background particles are occupying the MALTA2 word-encoding 1.6 ns window and that hit becomes dispalced or changed- decrease in hit efficiency.

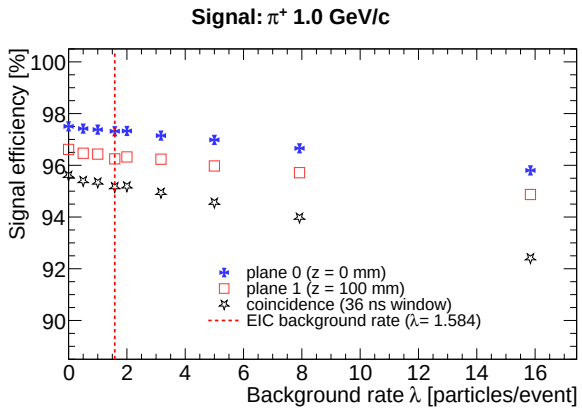


Fig. 10 Hit efficiency as a function of the background rate for 1 GeV pions. Per-plane efficiencies are shown for plane 0 and plane 1, together with the coincidence efficiency.

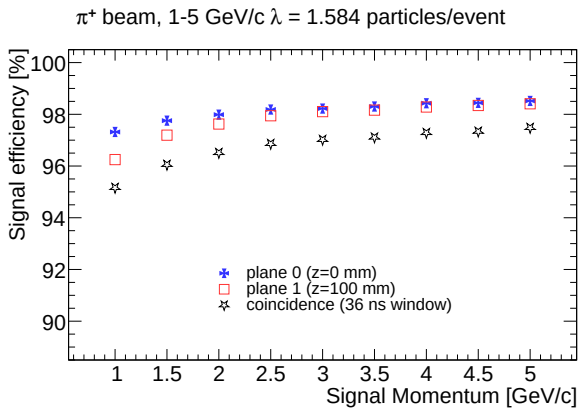


Fig. 11 Hit efficiency as a function of the signal momentum. Background rate is fixed at EIC background rate $\lambda = 1.584 \text{particles/event}$

5.6 Tracking efficiency, potential spatial/temporal resolution with background

This subsection can target the proper tracking efficiency using 2 FMT planes.

- Obtain hits per plane.
- Reconstruct a track from the two planes.
- Calculate pointing resolution of the track. In the simulation, signal is incident perpendicular onto the sensor. How is the direction reconstructed from the track?

5.7 Matching capability with background

6 Summary

Conclusions may be used to restate your hypothesis or research question, restate your major findings, explain the relevance and the added value of your work, highlight any limitations of your study, describe future directions for research and recommendations.

In some disciplines use of Discussion or 'Conclusion' is interchangeable. It is not mandatory to use both. Please refer to Journal-level guidance for any specific requirements.

Acknowledgment

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Appendix A: Appendix section

We consider a sequence of queueing systems indexed by n. It is assumed that each system is composed of J stations, indexed by 1 through J, and K customer classes, indexed by 1 through K. Each customer class has a fixed route through the network of stations. Customers in class k, $k = 1, \ldots, K$, arrive to the system according to a renewal process, independently of the arrivals of the other customer classes. These customers move through the network, never visiting a station more than once, until they eventually exit the system.

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